

Worksheet 2 - BDBS205 -

Database design and cloud computing

1. Given the table STUDENT(student_id, roll_no, name, email, phone, department_id)

Write SQL constraints for student table to:

- Make student_id auto increment
- Ensure email is unique
- Prevent NULL in name

2. Design an ER diagram for:

Hospital Management System

- A patient can consult multiple doctors.
- A doctor belongs to one department.
- A patient can have multiple appointments.
- Each appointment has date, time, and status.

3. a. Given tables Dept(dept_id PRIMARY KEY, dept_name) and Emp(emp_id PRIMARY KEY, emp_name, dept_id REFERENCES Dept(dept_id)), give the SQL query that lists each employee with the name of their department, including only employees who are assigned to an existing department?

3 b. Give the SQL query that finds the total salary paid by each department, returning department id and the sum of salaries from Emp(emp_id, emp_name, dept_id, salary)?

4. Explain the difference between a file-based system and a DBMS. Give two concrete problems that arise in a file-based system and how a DBMS solves them.
5. Define a view. Explain two practical advantages and one potential drawback of using views in a production system.